

Midpoint of a segment



1

a $M(5, -2)$

b $M(10, 6)$

2

a $M(5, 7)$

b $M\left(\frac{-11}{2}, -8\right)$

c $M(-3, 2)$

d $M(-3, -2)$

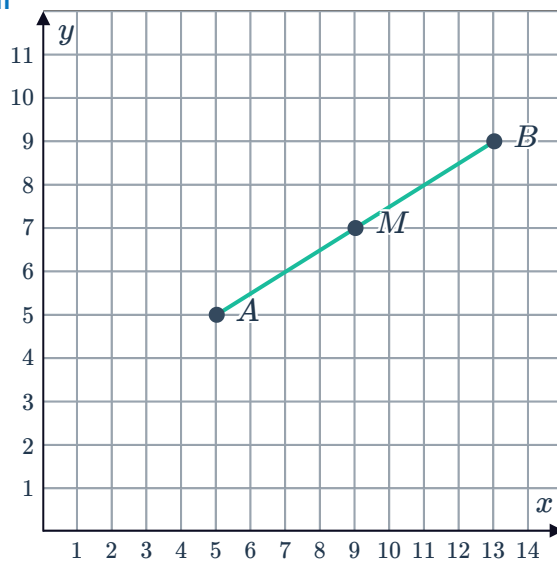
e $M(-2, 1)$

f $M(2, -4)$

3

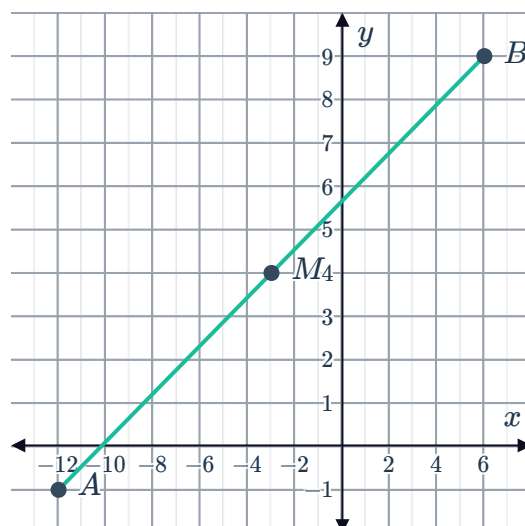
a i $M(9, 7)$

ii



b i $M(-3, 4)$

ii

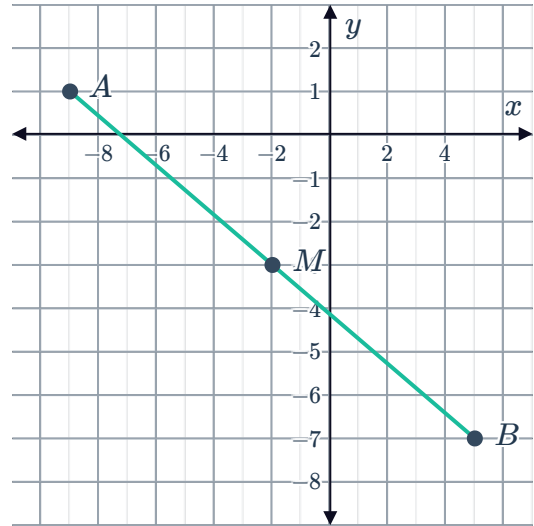


c

i $M(-2, -3)$



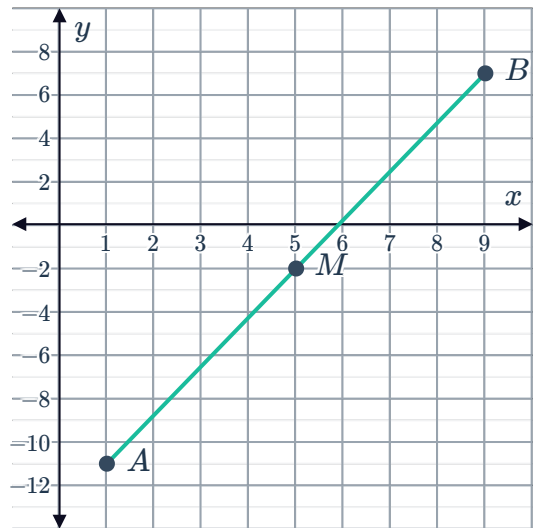
ii



d

i $M(5, -2)$

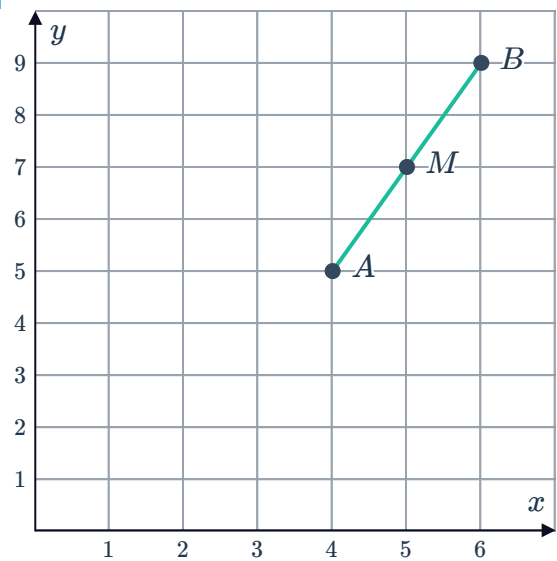
ii



e

i $M(5, 7)$

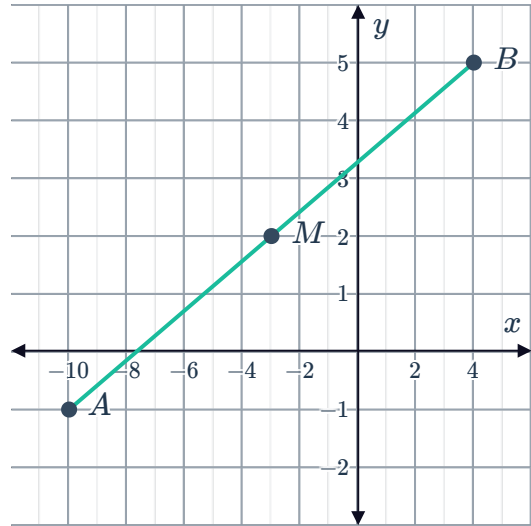
ii



f

i $M(-3, 2)$ 

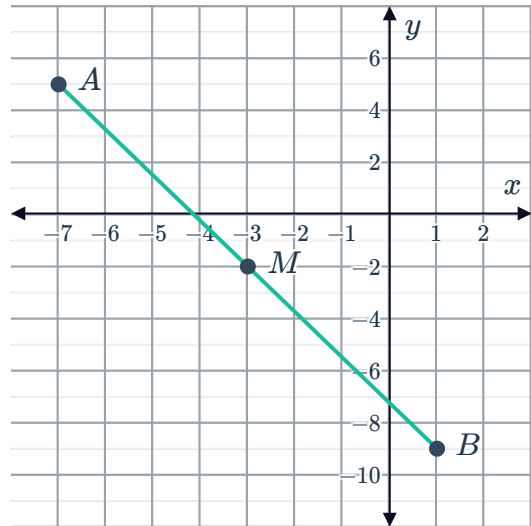
ii



g

i $M(-3, -2)$

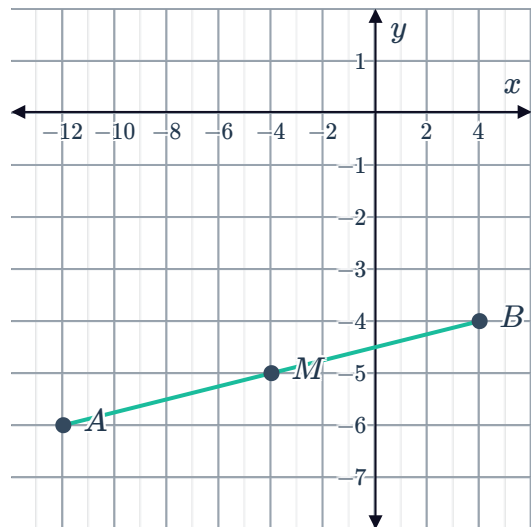
ii



h

i $M(-4, -5)$

ii



4

a $M(8, 6)$

b $M\left(\frac{7}{2}, \frac{1}{2}\right)$



c $M(-5, 9)$

d $M(6, -9)$

e $M(-9, -4)$

f $M(-5, 2)$

g $M(2, -6)$

h $M(-4, -7)$

i $M\left(-\frac{3}{4}, \frac{3}{4}\right)$

j $M\left(3, \frac{5}{2}\right)$

5 $(4m, 3n)$

6 $\left(m - 3, \frac{31n}{12}\right)$

7

a $p = -\frac{1}{4}$

b $q = 0$

8

a $M(3, 7)$

b $k = -11$

9 $Q(-6, 1), R(-8, 4), S(-10, 7)$

End points of a segment

10

a $A(5, 6)$

b $A(6, -5)$

c $A(5, 6)$

d $A(5, -6)$

e $A(-5, 6)$

f $A(4, -3)$

g $A(-4, -5)$

h $A(6, 22)$

i $A(-3, 3)$

j $A(-7, -4)$

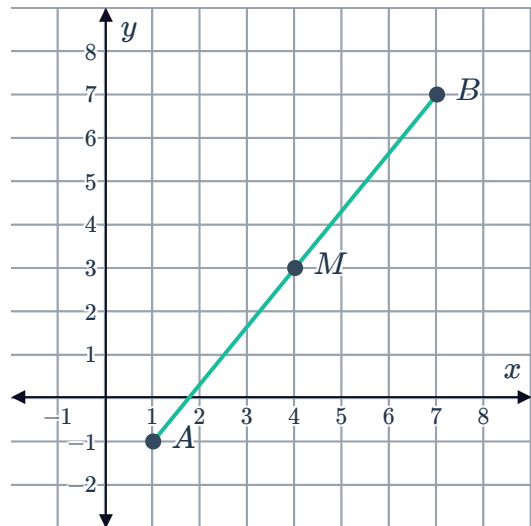
11

a

i $B(7,7)$



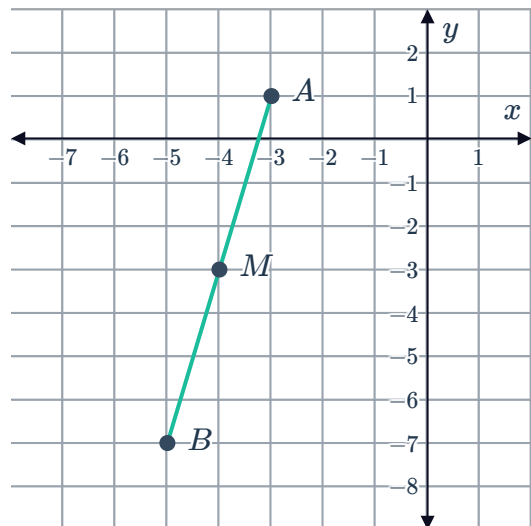
ii



b

i $B(-5, -7)$

ii



12 $B(8, -1)$

13

a Their diagonals bisect one another.

b

i $\left(-\frac{17}{2}, 3\right)$

ii $\left(-\frac{17}{2}, 3\right)$

c Yes

14

a $T(8, 8)$

b $S(6, 1)$

c Parallelogram, because the diagonals bisect each other.



Applications

15 29 million

16 63 years

17 $\left(\frac{69}{2}^{\circ}\text{N}, 103^{\circ}\text{E}\right)$

18
a 207 203

b 268 291